IMPROVED LOWER BOUNDS FOR SÁRKÖZY'S THEOREM IN FUNCTION FIELDS FROM INDEPENDENT SETS IN PRODUCTS OF GENERALIZED PALEY GRAPHS

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ABSTRACT. Let $P_{q,n}$ be the set of polynomials of degree $< n$ over $\mathbb{F}_q$ and let $k \geq 2$. Naslund proved that $P_{q,n}$ contains a subset of size $r_{k,2}(\mathbb{F}_q)^{n/2k}\, q^{n(1-1/k)}$ with no two elements differing by a nonzero k-th power, where $r_{k,2}(\mathbb{F}_q)$ is the independence number of the strong square of the generalized Paley graph $\mathrm{Paley}_k(\mathbb{F}_q)$, and instantiated this with $r_{k,2}(\mathbb{F}_q) \geq q$ and, for $k = 3$, with $r_{3,2}(\mathbb{F}_7) = 10$. We determine $r_{3,2}(\mathbb{F}_{13}) = 21$ by two independent exhaustive computations, which gives a set of size $|P_{13,n}|^{\,c}$ with $c = \frac{2}{3} + \frac{\log 21}{6 \log 13} = 0.864495\ldots$, the largest exponent this method has produced for cube differences and an improvement over Naslund's $q^{5n/6}$ at $q = 13$. We exhibit an infinite family of exactly computable cases: whenever the nonzero k-th powers of $\mathbb{F}_q$ form the multiplicative group of a subfield $\mathbb{F}_{q'}$ with $[\mathbb{F}_q : \mathbb{F}_{q'}] = d$, one gets sets of size $q^{n(1-1/dk)}$, which beats $q^{n(1-1/2k)}$ as soon as $d \geq 3$. Finally, explicit verified independent sets give $r_{4,2}(\mathbb{F}_{17}) \geq 41$ and $r_{7,2}(\mathbb{F}_{29}) \geq 139$, hence exponents $0.913841\ldots$ for fourth powers over $\mathbb{F}_{17}$ and $0.961815\ldots$ for seventh powers over $\mathbb{F}_{29}$, compared with 0.875 and $0.928571\ldots$ previously. All certificates are verified exhaustively by a standalone script included in the paper.

1. INTRODUCTION

Sárközy [11] proved that a subset of $\{1, \ldots, N\}$ of positive density contains two elements whose difference is a nonzero k-th power; the best quantitative bounds for the largest such set are $N^{1-o(1)}$ from above [8] and, from below, the constructions of Ruzsa [9] (improved for $k = 2$ by Lewko [5]), which give sets of size N^{1-c_k} with c_k close to $1/k$.

In the function field setting one replaces $\{1, \ldots, N\}$ by

$$P_{q,n} := \{f \in \mathbb{F}_q[T] : \deg f < n\}, \qquad |P_{q,n}| = q^n.$$

Green [3] showed, using the polynomial method of Croot–Lev–Pach and Ellenberg–Gijswijt, that a subset of $P_{q,n}$ containing no two distinct elements u, v with $u - v = b^k$ for some $b \in \mathbb{F}_q[T]$ has size at most $q^{n(1-c_{q,k})}$ for an explicit $c_{q,k} > 0$. Naslund [7] adapted Ruzsa's construction to this setting and proved the lower bound

$$|A| \;\geq\; r_{k,2}(\mathbb{F}_q)^{\,n/2k}\, q^{\,n(1-1/k)} \qquad (2k \mid n), \tag{1}$$

where $r_{k,2}(\mathbb{F}_q)$ denotes the independence number of the strong square of the generalized Paley graph $\mathrm{Paley}_k(\mathbb{F}_q)$ (definitions are recalled in Section 2). He proved $r_{k,2}(\mathbb{F}_q) \geq q$ whenever $\gcd(k, q-1) > 1$, giving sets of size $q^{n(1-1/2k)}$, and observed that for $k = 3$, $q = 7$ one has $\mathrm{Paley}_3(\mathbb{F}_7) = C_7$ and $r_{3,2}(\mathbb{F}_7) = \alpha(C_7 \boxtimes C_7) = 10$, giving the exponent

$$c_3(7) = \tfrac{2}{3} + \tfrac{\log 10}{6 \log 7} = 0.863882\ldots,$$

i.e. a cube-difference-free subset of $P_{7,n}$ of size $(5.371\ldots)^n = 7^{\,0.8638\ldots n}$. More generally, when $p = 2k + 1$ is prime, $\mathrm{Paley}_k(\mathbb{F}_p) = C_{2k+1}$ and Hales' formula $\alpha(C_{2k+1} \boxtimes C_{2k+1}) = k^2 + \lfloor k/2 \rfloor$ [4] applies. Beyond these cycle cases, the quantity $r_{k,2}(\mathbb{F}_q)$ was not computed in [7], and to our knowledge nowhere else. Subsequent work on Sárközy-type problems over $\mathbb{F}_q[T]$ (e.g. the extension of Green's upper bound to polynomials F with a number of roots divisible by p [10],

and bounds for differences of the form $P-1$ with P irreducible [2]) concerns upper bounds or other shift sets and does not affect the lower bound (1).

The purpose of this note is to evaluate $r_{k,2}(\mathbb{F}_q)$, exactly where we can and by explicit certificates otherwise, and to record the resulting improvements. Throughout, an *exponent* c means that $P_{q,n}$ contains a subset of size $\geq C\,q^{cn}$ with no nonzero k-th power difference, for all n and some $C = C(q,k) > 0$; by (1),

$$c_k(q) \;=\; 1 - \frac{1}{k} + \frac{\log r_{k,2}(\mathbb{F}_q)}{2k \log q} \tag{2}$$

is admissible.

Theorem 1.1. $r_{3,2}(\mathbb{F}_{13}) = 21$. *Consequently, for every n there is a subset of $P_{13,n}$ of size at least $13^{-6} \cdot 13^{\,cn}$ with no two elements differing by a nonzero cube, where*

$$c = \frac{2}{3} + \frac{\log 21}{6 \log 13} = 0.864495292331782279\ldots$$

This exceeds Naslund's exponent $5/6$ for $q = 13$ and also his value $c_3(7) = 0.863882\ldots$; it is the largest value of (2) for $k = 3$ among all prime powers $q \leq 73$ with $\gcd(3, q-1) > 1$, and the ϑ bound of Section 5 shows that no $q \in \{4, 7, 19, 25, 49, 64, 73\}$ can give more; for the remaining $q \leq 73$ our searches found nothing larger (Section 3), the closest case being $q = 16$ (Remark 3.1).

The exact value in Theorem 1.1 was obtained by two independent exhaustive computations (a branch-and-bound clique search with symmetry reduction, and a SAT solver proof that no independent set of size 22 exists), described in Section 3; the 21-element set is listed in Appendix A.

The next result is an infinite family where $r_{k,2}(\mathbb{F}_q)$ can be computed by hand and the resulting exponent is well above $1 - 1/2k$.

Theorem 1.2. *Let $q = q'^d$ with $\mathbb{F}_{q'} \subseteq \mathbb{F}_q$ a subfield, $d \geq 2$, and let k be a positive integer with $\gcd(k, q-1) = (q-1)/(q'-1)$, so that the set of nonzero k-th powers in $\mathbb{F}_q$ is exactly $\mathbb{F}_{q'}^{\times}$. Then $r_{k,2}(\mathbb{F}_q) = (q/q')^2$ and there are subsets of $P_{q,n}$ with no nonzero k-th power difference of size at least*

$$q^{-2k} \cdot q^{\,n(1-\frac{1}{dk})}.$$

For $d \geq 3$ this is larger than $q^{n(1-1/2k)}$, and it is consistent with the conjectural upper bound $q^{n(1-1/k^2)}$ of [7] since $k \geq d$.

For example $(q, q', k) = (8, 2, 7)$ gives exponent $20/21 = 0.952380\ldots$ for seventh powers over $\mathbb{F}_8$, against $13/14$ from (1) with $r_{7,2} \geq q$, and $(27, 3, 13)$ gives $38/39$ against $25/26$.

Finally, for $k = 4$ and $k = 7$ and moderate q we found large independent sets by local search; they are not claimed to be optimal, but each is an explicit certificate verified exhaustively.

Theorem 1.3. $r_{4,2}(\mathbb{F}_{17}) \geq 41$, $r_{4,2}(\mathbb{F}_{25}) \geq 49$, $r_{7,2}(\mathbb{F}_{29}) \geq 139$, $r_{5,2}(\mathbb{F}_{41}) \geq 128$, $r_{6,2}(\mathbb{F}_{37}) \geq 111$ *and* $r_{3,2}(\mathbb{F}_{16}) \geq 25$, *with the explicit sets of Appendix A and the ancillary file* `certificate.json`. *In particular there are subsets of $P_{17,n}$ without nonzero fourth power differences of size $\gg 17^{\,0.913841\ldots n}$ and subsets of $P_{29,n}$ without nonzero seventh power differences of size $\gg 29^{\,0.961815\ldots n}$.*

Table 1 summarises all values of $r_{k,2}(\mathbb{F}_q)$ we know, with upper bounds from the Lovász theta function. For $k = 5$ and $k = 6$ the cycle cases $q = 11$ and $q = 13$ of [7] remain the best we found; we verified $\alpha(C_{11} \boxtimes C_{11}) = 27$ and $\alpha(C_{13} \boxtimes C_{13}) = 39$ exactly with the same code.

TABLE 1. Independence numbers $r_{k,2}(\mathbb{F}_q) = \alpha(\mathrm{Paley}_k(\mathbb{F}_q)^{\boxtimes 2})$ and the exponent (2). "=" means an exact value (exhaustive computation, or a case of Theorem 1.2, or equality with the ϑ bound); "$\geq$" a verified certificate. The last column is $\lfloor \vartheta(\mathrm{Paley}_k(\mathbb{F}_q))^2 \rfloor$, an upper bound for $r_{k,2}(\mathbb{F}_q)$ (Section 5).

k	q	$\lvert K^\times \rvert$	$r_{k,2}(\mathbb{F}_q)$	exponent $c_k(q)$	ϑ^2 bound
3	7	2	$= 10$ [7]	0.863882443742489721	11
3	13	4	$= 21$ (Thm. 1.1)	**0.864495292331782279**	26
3	16	5	≥ 25	0.860160674573946862	36
4	17	4	≥ 41	**0.913841000294550348**	51
4	25	6	≥ 49	0.901132744390270945	57
5	11	2	$= 27$ [7, 4]	0.937447072997397867	29
5	41	8	≥ 128	0.930656687867260361	219
6	13	2	$= 39$ [7, 4]	0.952359778419282890	41
6	37	6	≥ 111	0.942020614408069151	127
7	8	1	$= 16$ (Thm. 1.2)	0.952380952380952381	16
7	29	4	≥ 139	**0.961815105020901528**	174

Proof status. Theorem 1.1: the lower bound $r_{3,2}(\mathbb{F}_{13}) \geq 21$ is verified by the standalone script of Appendix B; the upper bound $r_{3,2}(\mathbb{F}_{13}) \leq 21$ rests on two independent exhaustive computer searches (Section 3). Theorem 1.2 is proved by hand in Section 4. Theorem 1.3 is verified by the same script; no optimality is claimed there. The transfer statement (1) is Naslund's; we include a complete proof in Section 2 in the language used by our verifier.

2. The construction

Fix a prime power q and $k \geq 2$ with $\gcd(k, q-1) > 1$. Let

$$K = \{x^k : x \in \mathbb{F}_q\}, \qquad K^\times = K \setminus \{0\}, \qquad \widetilde{K} = K^\times \cup (-K^\times).$$

Note $-1 \in K$ whenever $(q-1)/\gcd(k, q-1)$ is even, in which case $\widetilde{K} = K^\times$; in general we symmetrise.

Definition 2.1. $\mathrm{Paley}_k(\mathbb{F}_q)$ is the graph on vertex set $\mathbb{F}_q$ in which $x \neq y$ are adjacent iff $x - y \in \widetilde{K}$. For graphs G, H the strong product $G \boxtimes H$ has vertex set $V(G) \times V(H)$, with $(g,h) \neq (g',h')$ adjacent iff $(g = g' \text{ or } gg' \in E(G))$ and $(h = h' \text{ or } hh' \in E(H))$. We write $r_{k,1}(\mathbb{F}_q) = \alpha(\mathrm{Paley}_k(\mathbb{F}_q))$ and $r_{k,2}(\mathbb{F}_q) = \alpha(\mathrm{Paley}_k(\mathbb{F}_q) \boxtimes \mathrm{Paley}_k(\mathbb{F}_q))$.

Thus $U \subseteq \mathbb{F}_q^2$ is independent in $\mathrm{Paley}_k(\mathbb{F}_q)^{\boxtimes 2}$ iff for all distinct $(x,y), (x',y') \in U$,

$$\text{not } \big(x - x' \in \widetilde{K} \cup \{0\} \ \text{ and } \ y - y' \in \widetilde{K} \cup \{0\}\big). \tag{3}$$

This is exactly the condition checked by the verifier of Appendix B. Since $\widetilde{K} \cup \{0\} \supseteq K$, an independent set in the symmetrised graph is a fortiori independent in the directed graph considered in [7], so the values in Table 1 are valid for both conventions; for all entries of the table $-1 \in K$ and the two graphs coincide.

For $f \in \mathbb{F}_q[T] \setminus \{0\}$ let $\deg f$ and $\operatorname{ord} f$ be the largest and smallest exponents with nonzero coefficient, and $\operatorname{lc} f$, $\operatorname{tc} f$ the corresponding coefficients. If $f = b^k$ with $b \neq 0$ then

$$k \mid \deg f, \quad \operatorname{lc} f = (\operatorname{lc} b)^k \in K^\times, \qquad k \mid \operatorname{ord} f, \quad \operatorname{tc} f = (\operatorname{tc} b)^k \in K^\times. \tag{4}$$

Theorem 2.2 (Naslund [7, Thm. 10])**.** *Let $U \subseteq \mathbb{F}_q^2$ be independent in $\mathrm{Paley}_k(\mathbb{F}_q)^{\boxtimes 2}$ and let $n = 2kN$. Then there is $A \subseteq P_{q,n}$ with $|A| = |U|^N q^{2(k-1)N}$ such that $u - v$ is not a nonzero k-th power for any $u, v \in A$. For general n one obtains $|A| \geq q^{-2k} |U|^{n/2k} q^{n(1-1/k)}$.*

Proof. Write $f = \sum_{i<n} c_i T^i$. The exponents $i \in [0, n-k]$ divisible by k are partitioned into the N pairs $\{i, n-k-i\}$ with $i \equiv 0 \pmod k$, $0 \leq i < n/2$ (note $n/2 \equiv 0 \pmod k$, so $i \leq n/2 - k$ and $n - k - i \geq n/2$). Let A consist of all f with $(c_{n-k-i}, c_i) \in U$ for each such i and arbitrary coefficients at all other exponents. Then $|A| = |U|^N q^{n-2N}$ and $n - 2N = 2(k-1)N$.

Let $u \neq v \in A$, $f = u - v$, and suppose $f = b^k$. Among the exponents j with coefficient $f_j \neq 0$ choose one maximising $|j - \frac{n-k}{2}|$, preferring $j > \frac{n-k}{2}$ in case of a tie (note $\frac{n-k}{2} \notin \mathbb{Z}$ when k is odd, and is not a multiple of k in any case, so $j \neq \frac{n-k}{2}$ whenever $k \mid j$).

Case $j > \frac{n-k}{2}$. Then $f_i = 0$ for $i > j$ and for $i < n-k-j$, so $\deg f = j$ and $\operatorname{ord} f \geq n-k-j$. By (4), $k \mid j$ and $f_j \in K^\times$; since $j \leq n-1$ and $k \mid n$ this forces $j \leq n-k$, so $j = n-k-i$ with $i := n-k-j \equiv 0 \pmod k$ and $i < n/2$, i.e. $\{i, j\}$ is one of the pairs and $(f_j, f_i) = (c_j, c_i) - (c'_j, c'_i)$ is the difference of two elements of U. Moreover either $f_i = 0$, or $\operatorname{ord} f = i$ and then $f_i \in K^\times$ by (4). Thus $f_j \in K^\times$ and $f_i \in K \cup \{0\} \subseteq \widetilde{K} \cup \{0\}$ with $(f_j, f_i) \neq (0,0)$, contradicting (3).

Case $j < \frac{n-k}{2}$. Now $f_i = 0$ for $i < j$ and for $i > n-k-j$, so $\operatorname{ord} f = j$ and $\deg f \leq n-k-j$. By (4), $k \mid j$ and $f_j \in K^\times$, and $j < n/2$, so $\{j, n-k-j\}$ is one of the pairs, with f_j in the second coordinate. Either $f_{n-k-j} = 0$ or $\deg f = n-k-j$ and $f_{n-k-j} \in K^\times$. Again $(f_{n-k-j}, f_j) \in (\widetilde{K} \cup \{0\})^2 \setminus \{(0,0)\}$ is a difference of two elements of U, contradicting (3). For general n apply this with $n' = 2k\lfloor n/2k \rfloor$ and $P_{q,n'} \subseteq P_{q,n}$. □

Remark 2.3. The argument only uses the outermost pair of coefficients at which u and v differ. One can group coefficients into blocks of length $s = mk$ and pair block t with block $n/s - 1 - t$; the same proof then shows that any set $U_m \subseteq (\mathbb{F}_q^s)^2$ which is independent in $L_m \boxtimes L_m$ is admissible, where L_m is the graph on $\mathbb{F}_q^s$ in which two distinct vectors are adjacent iff the highest index where they differ is divisible by k and the difference there lies in $\widetilde{K}$ (the reversed graph for the second factor is isomorphic to L_m by reversing coordinates). L_m is the iterated lexicographic product of copies of $\mathrm{Paley}_k(\mathbb{F}_q)$ and edgeless graphs E_q, and taking U_m to be a product of copies of U recovers Theorem 2.2. The exponent obtained is $\log_q \alpha(L_m^{\boxtimes 2})/(2mk)$, and $\alpha(L_m^{\boxtimes 2}) \geq r_{k,2}(\mathbb{F}_q)^m q^{2(k-1)m}$. Whether the inequality can be strict is a question about strong products of lexicographic products which we could not settle; for $q = 7$, $k = 3$, $m = 2$ the relevant subgraph $(C_7[C_7])^{\boxtimes 2}$ on 2401 vertices satisfies $\alpha \leq \alpha(C_7^{\boxtimes 4})$, and an extensive local search (Section 3) found nothing larger than the product value 100; the same happened for $(C_7[C_5])^{\boxtimes 2}$ and $(C_5[C_7])^{\boxtimes 2}$ (product value 50). We therefore do not use this refinement.

3. Computations

All graphs $\mathrm{Paley}_k(\mathbb{F}_q)^{\boxtimes 2}$ were generated from an explicit model of $\mathbb{F}_q$ ($\mathbb{F}_q = \mathbb{F}_p[t]/(\pi)$ with the lexicographically first monic irreducible π of degree r; elements are encoded as integers in $[0, q)$ by their base-p digit vectors). The encoding and the irreducible polynomial are stored in `certificate.json` so that the verifier can rebuild the field.

3.1. Exact value for $q = 13$, $k = 3$.

Here $K^\times = \{1, 5, 8, 12\} \ni -1$, so $\mathrm{Paley}_3(\mathbb{F}_{13})$ is the circulant $C_{13}(1, 5)$, 4-regular, with $r_{3,1}(\mathbb{F}_{13}) = 4$ (trivial to check). The strong square has 169 vertices and is 24-regular.

Method 1. A bitset branch-and-bound maximum clique search on the complement, with a greedy colouring bound (in the style of Tomita's MCQ). Because the graph is a Cayley graph on $\mathbb{F}_{13}^2$ we may assume that $(0,0)$ belongs to a maximum independent set, and the second vertex may be restricted to representatives of the orbits of the stabiliser of $(0,0)$ generated by

$(x,y)\mapsto(ax,y),(x,ay)$ $(a\in\widetilde{K})$, $(x,y)\mapsto(y,x)$ and the Frobenius map (10 orbits); after a representative v has been fully explored, its whole orbit is removed from the candidate set. The search proves $\alpha=21$ in $1.9\cdot10^6$ nodes (≈ 2 s). Without the orbit reduction it takes $5.2\cdot10^6$ nodes and gives the same value.

Method 2. An independent SAT encoding: one Boolean variable per vertex, a binary clause per edge, the unit clause fixing $(0,0)$, and a sequential-counter cardinality constraint "at least s true". CaDiCaL (via PySAT) finds a model for $s=21$ in 8 s and proves unsatisfiability for $s=22$ in 30 s.

Canaries. Method 1 returns $\alpha(C_7\boxtimes C_7)=10$, $\alpha(C_5\boxtimes C_5)=5$, $\alpha(C_{11}\boxtimes C_{11})=27$, $\alpha(C_{13}\boxtimes C_{13})=39$ and $\alpha(C_7[C_7])=9$, in agreement with [4] and with $\alpha(G[H])=\alpha(G)\alpha(H)$; the local search of the next subsection also recovers $\alpha(C_7^{\boxtimes3})=33$.

The 21-element set is listed in Appendix A. Together with Theorem 2.2 this proves Theorem 1.1; the comparison with other q is discussed next.

3.2. **Other q for $k=3$.** We ran an iterated local search for independent sets (perturbation by forced insertions, $(1,2)$-swap improvements, in the style of Andrade–Resende–Werneck [1]) on $\mathrm{Paley}_3(\mathbb{F}_q)^{\boxtimes2}$ for all prime powers $q\le73$ with $\gcd(3,q-1)>1$, i.e. $q\in\{4,7,13,16,19,25,31,37,43,49,61,64,67,7$ To beat the exponent of Theorem 1.1 one needs $r_{3,2}(\mathbb{F}_q)>q^{\log_{13}21}=q^{1.18693\ldots}$. The best sets found were

q	4	7	13	16	19	25	31	37	43	49	61	64	67	73
found	4	10	21	25	22	25	41	58	58	40	64	64	70	64
needed	6	11	22	27	33	46	59	73	87	102	132	140	148	163
$\lfloor\vartheta^2\rfloor$	4	11	26	36	27	25	59	96	109	65	138	64	170	148

For $q\in\{4,25,64\}$ the value found equals $\lfloor\vartheta^2\rfloor$ and is therefore exact (and equals q); for $q\in\{19,49,73\}$ the ϑ bound is below the threshold, so these q cannot beat $q=13$. For $q\in\{16,31,37,43,61,67\}$ the question is not settled by ϑ; the search values are well below the thresholds except for $q=16$.

Remark 3.1. $\mathrm{Paley}_3(\mathbb{F}_{16})$ is the 5-regular Clebsch graph ($K^\times$ is the subgroup of order 5 of $\mathbb{F}_{16}^\times$), with eigenvalues $5,1,-3$, so $\vartheta=6$ and $r_{3,2}(\mathbb{F}_{16})\le36$. Our search found $25=r_{3,1}(\mathbb{F}_{16})^2$ and nothing larger; $r_{3,2}(\mathbb{F}_{16})\ge27$ would give a larger exponent than Theorem 1.1. Neither the branch-and-bound nor the SAT approach settled this within our budget; we leave it open.

3.3. **$k\ge4$.** For $k=4$ Naslund's cycle case does not occur ($2k+1=9$ is not prime; $\mathrm{Paley}_4(\mathbb{F}_9)$ is $3K_3$ and $r_{4,2}(\mathbb{F}_9)=9$), so (1) gave only $7/8$. The search over $q\in\{5,9,13,17,25,29,41,49,53\}$ found the values $5,9,13,41,49,29,100,49,30$; only $q=17$ ($\mathrm{Paley}_4(\mathbb{F}_{17})=C_{17}(1,4)$, $r_{4,1}=6$) and $q=25$ improve on $7/8$, and $q=17$ is best. For $k=7$ the search over $q\in\{8,29,43\}$ found $16,139,150$; $q=29$ ($K^\times=\{1,12,17,28\}$, $r_{7,1}=11$) is best. For $k=5$ ($q\in\{11,16,31,41\}$: $27,16,70,128$) and $k=6$ ($q\in\{7,13,19,25,31,37\}$: $10,39,22,25,41,111$) the cycle cases $q=11$, $q=13$ remain best. The sets for $(k,q)=(4,17)$ and $(7,29)$ were re-found by several independent random seeds; the sets for $(5,41)$ and $(6,37)$ were the best of a handful of short runs (other seeds gave 127 and 100–101 respectively) and are probably not optimal. We do not claim optimality for any of the "$\ge$" entries beyond this search.

4. The subfield family

Proof of Theorem 1.2. $\mathbb{F}_q^\times$ is cyclic of order $q-1$, and the image of $x\mapsto x^k$ is its unique subgroup of order $(q-1)/\gcd(k,q-1)=q'-1$, which is $\mathbb{F}_{q'}^\times$. Hence $K=\mathbb{F}_{q'}^\times$ and $\widetilde{K}=\mathbb{F}_{q'}^\times$ (as $-1\in\mathbb{F}_{q'}$). Therefore $x,y\in\mathbb{F}_q$ are adjacent in $\mathrm{Paley}_k(\mathbb{F}_q)$ iff $x-y\in\mathbb{F}_{q'}^\times$, i.e. $\mathrm{Paley}_k(\mathbb{F}_q)$ is the disjoint union of the q/q' cliques given by the cosets of the additive subgroup $\mathbb{F}_{q'}$. Its strong square is

the disjoint union of $(q/q')^2$ cliques $K_{q'} \boxtimes K_{q'} = K_{q'^2}$, so $r_{k,2}(\mathbb{F}_q) = (q/q')^2$ exactly; an explicit independent set is $R \times R$ where R is a set of coset representatives of $\mathbb{F}_{q'}$ in $\mathbb{F}_q$. Theorem 2.2 gives sets of size $\geq q^{-2k}\big((q/q')^2\big)^{n/2k} q^{n(1-1/k)} = q^{-2k} q^{n(1-\log_q(q')/k)}$ and $\log_q q' = 1/d$. The comparison with $1 - 1/2k$ is $1/d < 1/2$. Finally $k \geq \gcd(k, q-1) = (q-1)/(q'-1) \geq q'^{d-1} \geq 2^{d-1} \geq d$, so $1 - 1/dk \leq 1 - 1/k^2$. □

For $(q, q', k) = (8, 2, 7)$ the set $R \times R$ with $R = \{0, t, t^2, t^2 + t\}$ (elements $0, 2, 4, 6$ in the encoding of Appendix A) is the certificate `k7_q8`. Since $\gcd(k, q-1)$ only depends on k modulo $q - 1$, each field $\mathbb{F}_q$ with a subfield of index ≥ 3 gives infinitely many exponents k.

5. Upper bounds and limits of the method

By Lovász [6], $\alpha(G) \leq \vartheta(G)$, $\vartheta(G \boxtimes H) = \vartheta(G)\vartheta(H)$, and for a d-regular graph on N vertices with least eigenvalue $\lambda_{\min}$ one has $\vartheta(G) \leq -N\lambda_{\min}/(d - \lambda_{\min})$. Hence

$$r_{k,2}(\mathbb{F}_q) \;\leq\; \vartheta(\mathrm{Paley}_k(\mathbb{F}_q))^2 \;\leq\; \Big(\frac{-q\lambda_{\min}}{|\widetilde{K}| - \lambda_{\min}}\Big)^2,$$

and the eigenvalues of the Cayley graph $\mathrm{Paley}_k(\mathbb{F}_q)$ are the character sums $\sum_{a \in \widetilde{K}} \psi(a)$, easily computed numerically. This gives the last column of Table 1 (values rounded down; they were computed in double precision from the eigenvalues of $\mathrm{Paley}_k(\mathbb{F}_q)$, the fractional parts are all at least 0.006, and for the cycle cases the exact value $\vartheta(C_p) = p\cos(\pi/p)/(1 + \cos(\pi/p))$ agrees). Naslund's Theorem 5 gives the weaker $r_{k,2} \leq q^{2-2/k}$ when $-1 \in K$; e.g. 30.57 versus 26.85 for $q = 13, k = 3$.

In terms of exponents, (2) can never exceed $1 - \frac{1}{k} + \frac{\log \vartheta(\mathrm{Paley}_k(\mathbb{F}_q))}{k \log q}$; for $q = 13$, $k = 3$ this ceiling is 0.8805, and for $q = 7$ it is 0.8721. Naslund's conjecture [7, Conj. 13] predicts that $1 - 1/k^2$ ($= 8/9$ for cubes) is the truth for $|A|$ in every $\mathbb{F}_q[T]$; the exponent 0.8645 of Theorem 1.1 is 0.024 below it.

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Appendix A. Certificates

Elements of $\mathbb{F}_p$ are residues; elements of $\mathbb{F}_{p^r}$ are encoded as $\sum_i a_i p^i$ for $\sum_i a_i t^i \in \mathbb{F}_p[t]/(\pi)$. For $\mathbb{F}_8$, $\pi = t^3 + t^2 + 1$; for $\mathbb{F}_{16}$, $\pi = t^4 + t^3 + 1$; for $\mathbb{F}_{25}$, $\pi = t^2 + t + 1$ (each the lexicographically first irreducible in the order used by the generator; the polynomial is recorded in `certificate.json`).

$r_{3,2}(\mathbb{F}_{13}) \geq 21$ ($K^\times = \{1,5,8,12\}$):

$$U = \{(0,12),(1,5),(1,9),(2,3),(2,7),(3,1),(3,10),(4,4),(5,8),(6,2),(6,11),(7,0),\\ (7,9),(8,3),(9,7),(10,1),(10,10),(11,4),(11,8),(12,2),(12,6)\}.$$

$r_{4,2}(\mathbb{F}_{17}) \geq 41$ ($K^\times = \{1,4,13,16\}$):

$$U = \{(0,12),(0,15),(1,3),(1,6),(1,9),(2,0),(2,11),(2,14),(3,2),(4,4),(4,7),(4,10),(5,1),(5,12),\\ (5,15),(6,6),(6,9),(7,4),(7,11),(7,16),(8,1),(8,13),(9,7),(9,10),(10,4),(10,16),(11,1),(11,6),\\ (11,13),(12,3),(12,8),(12,15),(13,0),(13,5),(14,11),(14,14),(15,3),(15,8),(16,0),(16,5),(16,10)\}.$$

The remaining sets (139 pairs for $q = 29$, $k = 7$, etc.) are in `certificate.json`.

Appendix B. Verification script

The following program (Python 3, standard library only) rebuilds each field from the data in `certificate.json`, computes K and $\widetilde{K}$ from scratch, checks (3) for every pair of elements of each certificate, and prints the exponent (2). Running `python3 verify_certificate.py certificate.json` prints `ALL CERTIFICATES VALID`. The ancillary files also contain the graph generator, the exact branch-and-bound solver, the local search, the SAT check and `reproduce.py`.

```
#!/usr/bin/env python3
"""verify_certificate.py -- standalone verifier (Python 3 standard library only).

For each entry of certificate.json it rebuilds F_q from the stated prime p,
degree r and monic irreducible polynomial, computes the set K of k-th powers,
and checks exhaustively that the listed set U of pairs (x,y) in F_q x F_q is an
independent set of the strong square of the (symmetrised) generalised Paley
graph Paley_k(F_q):  for all distinct (x,y),(x',y') in U it is NOT the case
that both x-x' and y-y' lie in K u (-K).   It then prints the resulting
exponent  c = 1 - 1/k + log_q(|U|)/(2k)  of Theorem 1 of the paper.

Usage:  python3 verify_certificate.py [certificate.json]
"""
import json, math, sys, itertools

def build_field(p, r, irr):
    """Elements are integers 0..p^r-1 encoding base-p digit vectors (low digit first)
    of coordinates in the basis 1, t, ..., t^{r-1} of F_p[t]/(irr)."""
    q = p ** r
    def tovec(i):
        v = []
        for _ in range(r):
            v.append(i % p); i //= p
        return v
    def fromvec(v):
        x = 0
        for c in reversed(v):
            x = x * p + (c % p)
        return x
    assert len(irr) == r + 1 and irr[-1] == 1, "irreducible must be monic of degree r"
    def add(a, b):
        return fromvec([(u + v) % p for u, v in zip(tovec(a), tovec(b))])
    def neg(a):
        return fromvec([(-u) % p for u in tovec(a)])
    def mul(a, b):
        A, B = tovec(a), tovec(b)
```

```
        prod = [0] * (2 * r - 1)
        for i, u in enumerate(A):
            for j, v in enumerate(B):
                prod[i + j] = (prod[i + j] + u * v) % p
        for d in range(2 * r - 2, r - 1, -1):
            c = prod[d]
            if c:
                for j in range(r + 1):
                    prod[d - r + j] = (prod[d - r + j] - c * irr[j]) % p
        return fromvec(prod[:r])
    # sanity: irreducibility check by verifying the multiplicative group has no zero divisors
    for a in range(1, q):
        for b in range(1, q):
            if mul(a, b) == 0:
                raise ValueError("polynomial is reducible: zero divisor found")
    return q, add, neg, mul

def verify(entry):
    p, r, k = entry["p"], entry["r"], entry["k"]
    q, add, neg, mul = build_field(p, r, entry["irreducible_low_to_high"])
    assert q == entry["q"]
    def power(x, e):
        y = 1
        for _ in range(e):
            y = mul(y, x)
        return y
    K = {power(x, k) for x in range(q)}          # k-th powers, including 0
    Ktil = (K | {neg(x) for x in K})              # K u -K  (contains 0)
    pairs = [tuple(v) for v in entry["pairs"]]
    assert len(set(pairs)) == len(pairs), "repeated pair"
    for (x, y) in pairs:
        assert 0 <= x < q and 0 <= y < q
    bad = 0
    for (x, y), (x2, y2) in itertools.combinations(pairs, 2):
        dx = add(x, neg(x2)); dy = add(y, neg(y2))
        if dx in Ktil and dy in Ktil:
            bad += 1
    size = len(pairs)
    c = 1 - 1 / k + math.log(size) / (2 * k * math.log(q))
    return q, k, size, bad, c, len(K) - 1

if __name__ == "__main__":
    fn = sys.argv[1] if len(sys.argv) > 1 else "certificate.json"
    certs = json.load(open(fn))
    allok = True
    for name, entry in certs.items():
        q, k, size, bad, c, nK = verify(entry)
        ok = (bad == 0)
        allok &= ok
        print(f"{name:8s} q={q:3d} k={k} |K*|={nK:3d} |U|={size:4d} conflicts={bad} "
              f"{'OK ' if ok else 'FAIL'} exponent >= {c:.9f}  (Naslund: {1-1/(2*k):.9f})")
    print("ALL CERTIFICATES VALID" if allok else "SOME CERTIFICATE FAILED")
    sys.exit(0 if allok else 1)
```